

Waseem Irfan

Gujranwala, Pakistan

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EDUCATION

NUST College of Electrical & Mechanical Engineering

Bachelor of Mechatronics Engineering

November 2021 – June 2025

Rawalpindi, Pakistan

EXPERIENCE

Powersoft19

September 2025 – Present

Embedded Systems Developer

Lahore, Pakistan

- Architected and deployed the master control firmware for a bi-directional power dock by integrating the TI C2000 (DSP) platform for efficient DC/AC inversion and closed-loop charge/discharge regulation. Implemented safety-critical pack-dock communication, real-time telemetry handling, protection interlocks, and BLE-based OTA updates for dock, pack, and C2000 firmware.
- Developed a Battery Management System (BMS) on STM32H755ZITx featuring cell monitoring via LTC6804-2, SoC estimation, protection mechanisms, configurable parameters, cell balancing and multi-protocol communication (Ethernet, iso-SPI, CAN, UART) for reliable and intelligent battery pack control.

Cowlar Design Studio (backed by Y-Combinator)

July 2023 – October 2024

Embedded Design Engineer

Islamabad, Pakistan

- Developed an ESP32-S3 based kitchen dashboard to automate restaurant operations, deployed in Qatar, optimizing order management, reducing processing time by 30%, enhancing workflow, and cutting operational costs by 45%.
- Designed an automotive diagnostic tool (Ignition coil tester) using the PADAUK PMC232 microcontroller to evaluate vehicle ignition coils, distinguishing functional and faulty coils with precision.

PROJECTS

Modular Self-Reconfigurable Robots (Link) | Final Year Design Project

June 2025

- Developed ESP32-based Modular Robots, BU04-Kit for indoor positioning, MPU6050 for orientation sensing, Li-Po cells with BMS for power management, N20 encoder motors with PID control, and magnetic docking for self-reconfiguration.

Kitchen Display System (Link) | FreeRTOS, MQTT, UART, LVGL, ESP-IDF

August 2024

- Developed an ESP32-S3 VGA based kitchen dashboard with real-time order display, hardware navigation, and status updates, integrating a thermal printer and buzzer alert system to optimize workflow.

Warehouse Management Robot (Link) | Arduino, SolidWorks, 3D Printing

May 2024

- Engineered an autonomous pick-and-place robot for the NERC 2024, featuring a 5-DoF robotic mechanism that scans a grid map to pick and place multiple objects in parallel, achieving 6x faster performance and significantly boosting productivity and operational efficiency.

H-Bridge Motor Driver (Link) | Altium Designer, Proteus

November 2023

- Crafted an H-Bridge for robot's drive train, enabling reliable 24V operation with a 10A stall current capacity, improving motor control efficiency and system performance.

TECHNICAL SKILLS

PCB Design: Altium Designer | Proteus | PDN Analyzer

Industrial Automation: CNC (G-code, M-code) | PLC (Ladder Logic)

Programming: C/C++ | Assembly | Python | Verilog | FreeRTOS | DSA & OOP

Embedded Boards: BU04-Kit | STM32 | ESP32-S3 | AT89C51 | Raspberry Pi | Radxa ZERO 3W

Communication Protocols: I2C | SPI | WIFI | UWB | BLE | MQTT | HTTP | SNMP | ESP-NOW

Frameworks & Tools: ESP-IDF | STM32CubeIDE | LabVIEW | MATLAB | CMake | Git | ROS2 | Linux(Ubuntu)

AWARDS & CERTIFICATIONS

- Participated in COMPPEC 2025, NUST CEME (Link)
- Best Product Design Award - Hackathon 2023, NUST CEME (Link)
- Participated in NUST Robotics & Automation Club (Link)
- Microcontroller Embedded C Programming: Absolute Beginners - Udemy (Link) Completed
- Mastering Microcontroller and Embedded Driver Development - Udemy (Link) Completed
- Advanced Embedded Linux Development Specialization - Coursera (Link) In Progress

Soft Skills: Strategic Leadership, Project Execution and Planning, Effective Communication, Analytical Problem Solving